

PROMPTMINER-SAM2: PROMPT MINING AND UNCERTAINTY-GUIDED REFINEMENT FOR OVERHEAD BUILDING INSTANCE SEGMENTATION

Anonymous Author(s)

Anonymous Affiliation(s)

Abstract—Overhead building instance segmentation is challenged by visually similar adjacent roofs, shadows, and narrow inter-building gaps. Two-stage detectors localize candidate instances, but a proposal box alone gives a promptable mask decoder little information about the target shape. We propose PromptMiner-SAM2, which maps proposal-conditioned features into SAM2’s prompt interface. An explicitly supervised coarse shape yields geometrically aligned point, box, and dense-mask prompts. To improve masks at ambiguous locations, an uncertainty-guided decoder-tail refiner updates low-confidence output locations. Experiments on the WHU dataset and NWPU VHR-10 benchmarks show that PromptMiner-SAM2 achieves 73.4% mask AP on the WHU dataset and 67.8% mask AP on NWPU VHR-10. Ablations further show that combining the three prompt types with uncertainty-guided refinement improves high-IoU mask quality.

Index Terms—Instance segmentation, remote sensing, SAM2, prompt learning, building extraction

I. INTRODUCTION

Building instance segmentation from overhead imagery supports cadastral mapping, urban planning, and post-disaster assessment [18, 19]. The task is more demanding than semantic building extraction: a model must separate adjacent structures and recover each footprint despite small roofs, shadows, weak contrast, and narrow gaps [18, 19]. Two-stage instance segmenters address the *where* question by generating object proposals, but the proposal box alone supplies little information about which pixels inside that extent belong to the target instance [8, 10].

Promptable foundation models perform mask decoding conditioned on prompts. Segment Anything (SAM) accepts sparse point and box prompts as well as dense mask prompts [1], and SAM2 retains this prompt-conditioned decoding paradigm while providing hierarchical visual representations [2]. A related effort has also scaled SAM-based segmentation to remote-sensing imagery [3]. A direct detector–SAM pipeline can use each proposal as a box prompt. Yet box prompting leaves the decoder to resolve

the target shape and to exclude nearby instances from the same region. This limitation is especially acute in overhead scenes with tightly packed buildings, where a proposal contains more instance-specific information than its rectangular extent alone. How to translate this proposal information into standard SAM2 prompts remains insufficiently explored.

Existing automatic-prompt approaches address related but different problems. RSPrompter learns category-related prompt embeddings from multi-scale features for remote-sensing instance segmentation [4]. AoP-SAM predicts prompt candidates and uses adaptive sampling and filtering to automate prompt selection efficiently [5]. PA-SAM inserts a trainable prompt adapter inside the SAM mask decoder to enrich sparse and dense *prompt features* with image details and mined hard points [6]. These methods learn prompt representations, select prompt candidates, or adapt decoder-side prompt features. SAMRefiner takes a different route by excavating point, box, and mask prompts from a pre-existing coarse mask for post-hoc refinement [7]. However, none represents the spatial information tied to each detector proposal as geometrically aligned point, box, and dense-mask prompts for SAM2. This leaves an open connection between proposal-local shape information and promptable mask decoding in crowded overhead imagery.

We propose *PromptMiner-SAM2*, a two-stage model that couples a SAM2-based detector with SAM2 prompt-conditioned mask decoding. Hierarchical features form a detection pyramid for proposal discovery, while the SAM2 image embedding and high-resolution decoder features are used for final mask prediction. For each proposal, an explicitly supervised coarse shape yields foreground–background points and a dense-mask prompt; the proposal itself supplies the box prompt.

Our contributions are threefold:

- We adapt SAM2 to a two-stage instance-segmentation network by repurposing its hierarchical Hierarchical features as a PAFPN for RPN/RoI instance discovery, while retaining the SAM2 prompt encoder and mask decoder for final segmentation.
- We introduce CSPM, which converts a proposal-level coarse mask into aligned point, box, and dense-mask

inputs for SAM2 decoding.

- We introduce an uncertainty-guided decoder-tail point refiner (UDPR) that directly interacts with SAM2 decoder logits and upscaled features to selectively calibrate low-confidence locations for localized mask refinement.

II. METHOD

A. Overall Architecture

PromptMiner-SAM2 couples a two-stage detector with SAM2’s prompt-to-mask interface. Hierarchical features support PAFPN/RPN/RoI proposal discovery, and CSPM transforms each proposal-local coarse shape into point, box, and dense prompts for adjacent buildings.

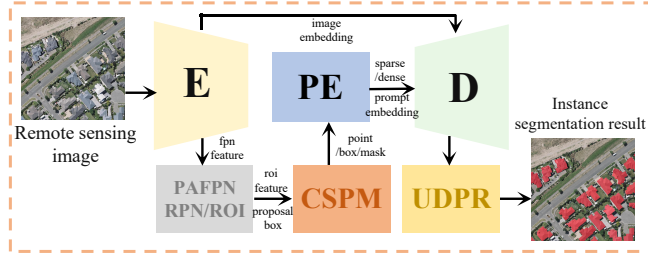


Fig. 1. Overall Architecture of PromptMiner-SAM2

Each proposal b_r anchors its multi-level RoIAlign feature X_r and all prompts in an image-aligned prompt coordinate frame. SAM2 decodes the image embedding with these proposal-specific prompts, and the resulting upscaled features enter UDPR. Points encode interior/exterior cues, the box constrains extent, and the dense prompt supplies a local shape prior.

B. Coarse-Shape Prompt Mining (CSPM)

For each proposal, the coarse branch maps its RoI feature X_r to a proposal-local shape logit map

$$C_r = f_{\text{coarse}}(X_r), \quad C_r \in \mathbb{R}^{1 \times 64 \times 64}. \quad (1)$$

Figure 2 summarizes the conversion of the coarse shape into the three SAM2 prompt types. Let $p_r = \sigma(C_r)$. PromptMiner derives a foreground support at probability 0.5 from p_r , while retaining the signed logits C_r for the dense prompt. It mines two positive and two negative points (2P2N) from this shared coarse shape. The first positive point is selected among confident foreground interior pixels ($p_r \geq 0.6$) by its foreground confidence weighted by its normalized distance to the support boundary; the second follows the same criterion while remaining spatially separated from the first. The first negative point comes from the one-pixel outer ring with low foreground probability ($p_r \leq 0.3$). The second is selected from

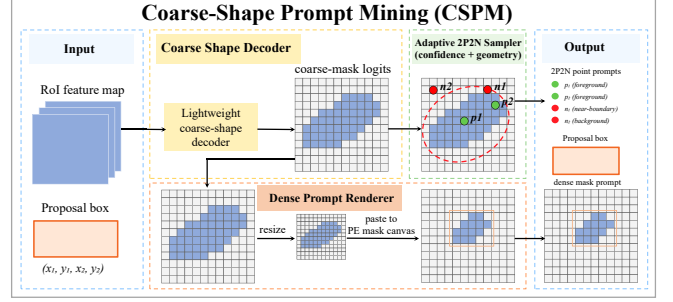


Fig. 2. Coarse-shape prompt mining (CSPM).

background outside a dilation of the foreground support and is separated from the first positive and ring-negative points.

This construction assigns distinct geometric roles to the four points. The positive pair anchors the object interior without repeatedly sampling one local peak. In contrast, the ring negative is tied to the immediate exterior of the coarse shape, whereas the second negative represents a less local background context. The 2P2N set records the coarse shape as sparse signed point prompts before they are transferred from the RoI frame to the image frame.

The resulting coordinates and positive/negative labels are mapped from the RoI frame to image coordinates. In parallel, the signed coarse logits are resized to the support of b_r and pasted onto the mask-input canvas:

$$S_r = \mathcal{M}(C_r, b_r), \quad Q_r = \text{Place}(C_r, b_r), \quad (2)$$

Here, $\mathcal{M}$ denotes 2P2N selection and coordinate mapping, and Place is the resize-and-paste operation. The two corners of b_r provide the box prompt. SAM2 jointly encodes the point, box, and dense-mask prompts, then decodes them with the image embedding in one pass:

$$Y_r^0 = \mathcal{D}(\mathcal{E}_I(I), \tilde{\mathcal{E}}_P(S_r, b_r, Q_r)), \quad (3)$$

where $\mathcal{E}_I$, $\tilde{\mathcal{E}}_P$, and $\mathcal{D}$ denote the SAM2 image encoder (including its decoder-side high-resolution features), complete SAM2 prompt-embedding pathway, and mask decoder, respectively. The pathway $\tilde{\mathcal{E}}_P$ includes the SAM2 composition of the dense-mask embedding. Q_r retains the signed rather than binarized coarse logits. Thus, the dense prompt preserves the spatial variation of the proposal-local shape, while the point and box prompts provide discrete geometric constraints to the same decoder.

C. Uncertainty-Guided Decoder-Tail Refinement (UDPR)

The same decoding pass also returns an upscaled decoder feature map U_r and the first output mask token t_r , in addition to the decoder logit map Y_r^0 . On the raster grid $\mathcal{G}_r$, UDPR visits the K_r locations with the smallest logit magni-

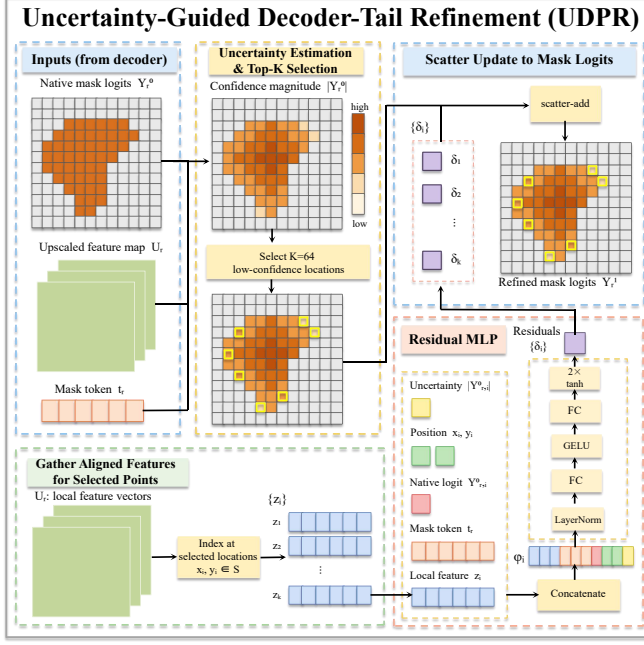


Fig. 3. Uncertainty-guided decoder-tail refinement (UDPR).

tudes, as illustrated in Figure 3:

$$\begin{aligned} K_r &= \min(K, |\mathcal{G}_r|), \\ \Omega_r &= \text{BottomK}(|Y_r^0|, K_r), \quad K = 64, \end{aligned} \quad (4)$$

where BottomK returns the K_r indices in $\mathcal{G}_r$ with the smallest logit magnitudes. For each $i \in \Omega_r$, the aligned feature z_i is gathered from U_r and combined with the token, normalized location, local logit, and logit magnitude:

$$\phi_i = [z_i, t_r, Y_{r,i}^0, x_i, y_i, |Y_{r,i}^0|]. \quad (5)$$

The local feature describes the decoder output near i , the mask token conditions the update on the decoded instance, and the coordinate and logit terms specify the position and current decision state. A lightweight MLP g then predicts a bounded residual for each selected location:

$$\delta_i = 2 \tanh g(\phi_i), \quad i \in \Omega_r, \quad (6)$$

The refined logit map is formed by scattering these residuals back to their original positions:

$$Y_{r,i}^1 = \begin{cases} Y_{r,i}^0 + \delta_i, & i \in \Omega_r, \\ Y_{r,i}^0, & i \notin \Omega_r. \end{cases} \quad (7)$$

This selective update leaves all nonselected decoder logits unchanged and uses the outputs of the single prompt-conditioned decoding pass throughout.

III. EXPERIMENTS

A. Training Details

Our implementation uses PyTorch and distributed data parallelism on four NVIDIA RTX 3090 GPUs. Each GPU processes one image and gradients are accumulated for two iterations, resulting in an effective global batch size of eight. We train with automatic mixed precision and AdamW (initial learning rate 5×10^{-4} , weight decay 0.05, and $\epsilon = 10^{-6}$). The learning rate is linearly warmed up for the first 100 optimizer steps from 0.1% of its initial value, followed by cosine annealing to 0.1% of that value. Training uses cross-entropy, Smooth L1, binary cross-entropy, and Dice losses.

B. Evaluation Protocol and Comparisons

We evaluate on the WHU dataset [20], a single-class building benchmark with a 2,943/627/2,220 train/validation/test split, and NWPU VHR-10 [21], a ten-class benchmark. Since NWPU VHR-10 originally provides detection annotations only, we use the publicly released instance masks of Su *et al.* [22] and its 520/130 train/evaluation split. We use COCO-style bbox and mask AP, AP₅₀, and AP₇₅. Table 1 reports the comparative results under this evaluation protocol. To assess the proposed method, we compare it with 11 representative instance-segmentation methods. These methods fall into two broad groups: CNN-based methods, including two-stage, one-stage, and cascade designs; and methods that exploit global sequence modeling, including transformer-style mask prediction and foundation-model prompting. This comparison includes conventional instance-segmentation pipelines and methods with global sequence modeling.

Quantitatively, PromptMiner-SAM2 obtains the highest mask AP on both benchmarks. On the WHU dataset, it reaches 73.4% mask AP and 85.5% mask AP₇₅, exceeding the strongest competing results by 0.1 and 1.6 points, respectively. Its bbox AP of 76.4% is also competitive, only 0.3 points below the best result. On NWPU VHR-10, our method achieves the best bbox AP (72.8%), bbox AP₇₅ (84.0%), mask AP (67.8%), mask AP₅₀ (95.4%), and mask AP₇₅ (74.0%).

Figure 4 provides qualitative comparisons on the two benchmarks. In densely built or visually ambiguous areas, where shadows, weak roof contrast, and narrow inter-building gaps complicate instance delineation, PromptMiner-SAM2 recovers more complete building footprints and produces cleaner separations between adjoining instances. These visual differences are consistent with its stronger high-IOU mask accuracy in Table 1.

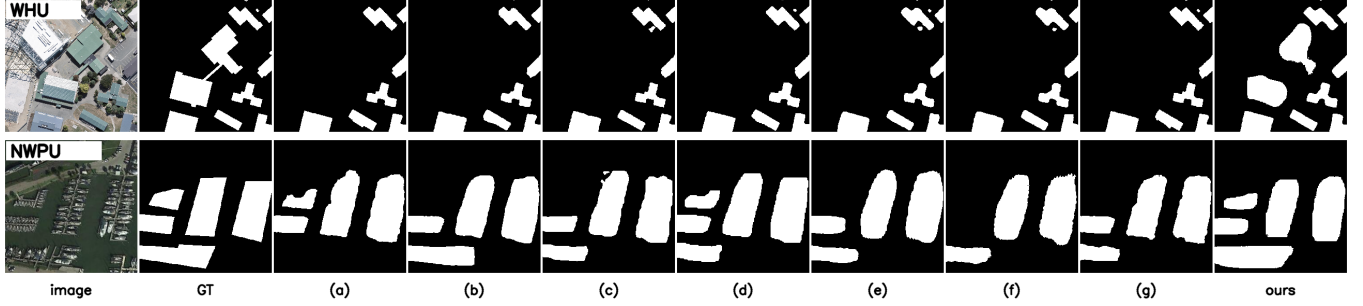


Fig. 4. Qualitative comparison on the WHU and NWPU VHR-10 datasets. From left to right, the columns show the input image, ground truth, predictions of (a) MaskDINO, (b) RSPrompter, (c) CondInst, (d) YOLO11, (e) HTC, (f) Mask R-CNN, (g) M2F, and PromptMiner-SAM2 (ours).

Table 1. Comparative analysis on the WHU and NWPU VHR-10 datasets. Each entry reports AP / AP₅₀ / AP₇₅ (%) for boxes or masks.

Type	Method	WHU						NWPU VHR-10					
		Bbox (%)			Mask (%)			Bbox (%)			Mask (%)		
		AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
<i>CNN-based</i>	Mask R-CNN [8]	70.9	89.9	80.9	68.2	90.0	80.0	50.1	82.1	57.7	49.1	81.7	50.5
	MS R-CNN [9]	68.8	89.4	80.0	67.7	89.6	79.5	51.5	85.2	59.2	52.7	82.6	56.8
	HTC [10]	72.7	90.6	82.5	69.2	90.7	81.6	58.9	86.9	67.7	51.3	84.6	53.3
	CondInst [11]	74.2	90.0	82.4	69.7	90.2	80.9	65.0	90.1	79.6	62.8	90.7	68.1
	RTMDet-Ins-S [14]	64.2	85.6	75.3	59.8	85.8	71.2	59.6	91.5	70.6	46.3	79.2	47.2
	YOLO11s-seg [15]	76.7	91.2	85.4	66.8	91.3	79.1	71.0	94.4	79.5	63.9	93.9	67.3
<i>Global sequence modeling</i>	Mask2Former [13]	61.4	82.7	70.7	64.5	84.1	75.6	60.3	77.9	66.8	61.5	84.9	67.8
	MaskDINO [16]	73.3	90.8	82.7	73.3	90.3	82.5	65.7	88.9	74.2	65.8	89.6	71.5
	RS4D-box [17]	64.5	87.6	74.7	64.1	87.8	75.2	57.0	82.5	65.4	54.3	80.1	57.6
	RSIISN [18]	74.1	90.0	84.4	72.1	90.1	83.9	58.1	89.3	66.8	62.9	89.7	67.1
	RSPrompter [4]	72.5	91.0	81.7	72.5	92.0	82.9	70.3	93.6	81.0	66.1	92.7	70.6
Ours	PromptMiner-SAM2	76.4	91.5	85.4	73.4	91.4	85.5	72.8	93.9	84.0	67.8	95.4	74.0

C. Ablation Study

Table 2 reports the effect of prompt composition and decoder-tail refinement on NWPU VHR-10. All variants use the same data split, initialization, training schedule, and checkpoint-selection rule. Starting from point prompts (P), adding the proposal box ($P+B$) raises mask AP from 63.0% to 65.3% and mask AP₇₅ from 67.4% to 71.7%, respectively. This 2.3-point mask-AP gain, obtained without a corresponding bbox gain, shows that the box primarily resolves instance extent for mask decoding rather than improving proposal localization. Adding the dense-mask prompt ($P+B+M$) further raises mask AP to 65.9% and bbox AP to 72.1%. Thus, the dense prompt transfers coarse shape information beyond the discrete point and box prompts, although its isolated improvement at AP₇₅ is modest.

Adding UDPR to $P+B+M$ raises mask AP/AP₇₅ from 65.9/71.5% to 67.8/74.0%, and increases bbox AP₇₅ by 2.9 points. Across the complete path, the gains over P reach 4.8 mask-AP and 6.6 AP₇₅ points, showing that CSPM improves shape conditioning while UDPR strengthens high-IoU masks.

Table 2. Prompt-path ablation on NWPU VHR-10. P , B , and M denote point, box, and dense-mask prompts.

Variant	Bbox (%)			Mask (%)		
	AP	AP ₅₀	AP ₇₅	AP	AP ₅₀	AP ₇₅
P	71.8	93.6	81.9	63.0	92.6	67.4
$P+B$	71.7	93.5	81.7	65.3	93.5	71.7
$P+B+M$	72.1	94.4	81.1	65.9	93.5	71.5
$P+B+M$ + UDPR	72.8	93.9	84.0	67.8	95.4	74.0

IV. CONCLUSION

PromptMiner-SAM2 couples two-stage proposals with SAM2 through coarse-shape prompts and UDPR. CSPM transforms each proposal-local shape into aligned point, box, and dense-mask prompts in SAM2’s native interface, reducing ambiguity between adjacent buildings. UDPR selectively updates low-confidence decoder logits to improve boundary-sensitive masks while preserving confident regions. The resulting design achieves strong performance on both benchmarks.

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